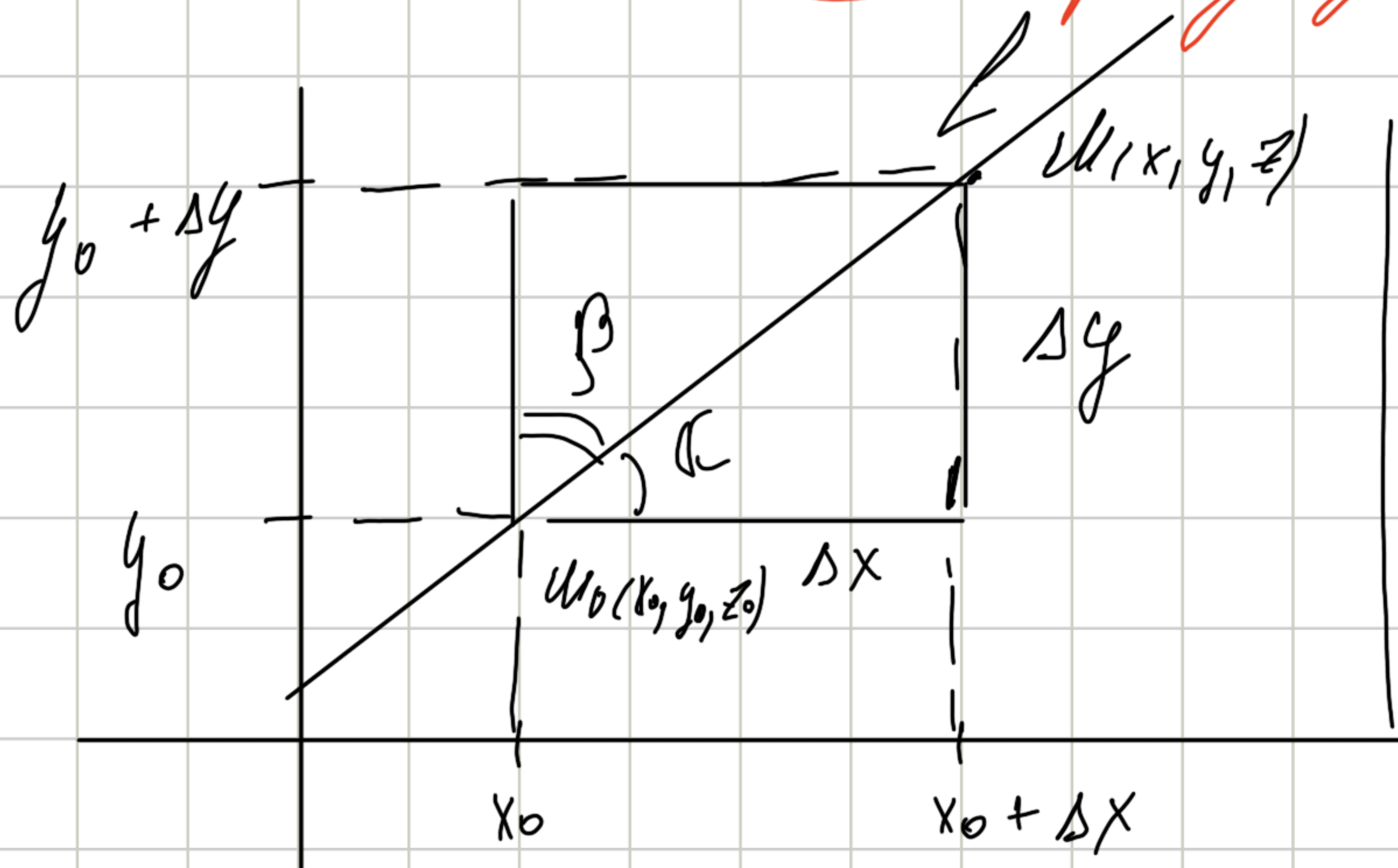


§8 Пространство по направлению.



$u(x, y) \in D$
 $u_0(x, y), z_0 = F(u_0)$
 направление
 $\vec{l} = (\cos \alpha, \cos \beta)$

$$u_0(x_0 + \Delta x, y_0 + \Delta y) \in \mathcal{L}$$

$$|\vec{l}| = \sqrt{\cos^2 \alpha + \cos^2 \beta} =$$

$$= \sqrt{\cos^2 \alpha + \sin^2 \alpha} = 1$$

$$\lim_{u \rightarrow u_0} \frac{F(u) - F(u_0)}{|\vec{l}|} = \frac{\partial F}{\partial \vec{l}}$$

- производная от функции F по направлению $\vec{l}$

$$x = x_0 + \|u\| \cos \alpha = x_0 + C \cdot \cos \alpha$$

$$y = y_0 + \|u\| \cos \beta = y_0 + C \cdot \cos \beta$$

$$\frac{F(u) - F(u_0)}{C} = \frac{F(x_0 + C \cdot \cos \alpha, y_0 + C \cdot \cos \beta) - F(x_0, y_0)}{C} =$$

$$= \frac{F(C) - F(0)}{C - 0} = \frac{\Delta F}{\Delta C}$$

$$\frac{\partial F}{\partial C} = \frac{d F(C)}{d C} = \frac{d}{d C} F(x_0 + C \cdot \cos \alpha, y_0 + C \cdot \cos \beta) =$$

$$= \frac{\partial F}{\partial x} \frac{dx}{dC} + \frac{\partial F}{\partial y} \frac{dy}{dC} = \frac{\partial F}{\partial x} \cos \alpha + \frac{\partial F}{\partial y} \cos \beta$$

$$\frac{\partial F}{\partial l} = \frac{\partial F}{\partial x} \cos \alpha + \frac{\partial F}{\partial y} \cos \beta, \quad \vec{l} = l \cos \alpha, \cos \beta$$

при 3 направлениях: $\frac{\partial F}{\partial l} = \frac{\partial F}{\partial x} \cos \alpha + \frac{\partial F}{\partial y} \cos \beta + \frac{\partial F}{\partial z} \cos \gamma \quad (1)$

$$\vec{l} = l \cos \alpha, \cos \beta, \cos \gamma, \quad \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$